Memory and chaos in a physics-constrained relaxation substrate: phase diagram, multi-timescale forgetting, and disturbance robustness

[Author Name]*

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Abstract

Reservoir computing exploits the fading-memory dynamics of a physical substrate, yet the memory–chaos trade-off is usually studied on abstract recurrent networks with hand-tuned leak rates. We characterize a physics-constrained relaxation substrate inspired by Si_3N_4 shallow-trap charge storage: N units with a log-normal time-constant spectrum (median $\tau_0 \approx 174\,\mu$s, CV $= 0.20$) coupled through a per-pulse, topology-dependent modulation of the injection coefficient. Sweeping the coupling strength κ across five topology families and measuring the finite-time Lyapunov exponent λ, the held-out Jaeger memory capacity, and input separation, we find a sharp order–chaos transition at $\kappa^* \in (25, 30)$, with the held-out memory capacity peaking 24–53% above the uncoupled baseline just before the transition, and deep chaos destroying memory. The decay of linear memory follows an analytically derived forgetting kernel $M(t) = \int p(\tau)e^{-t/\tau}\,d\tau$ over the log-normal trap spectrum (Pearson $r = 0.97$ against the measured memory-capacity curve): the $1/e$ horizon is pinned near $\tau_0/\bar{\Delta t} \approx 16$ pulses regardless of spectrum width, while the width CV controls the tail weight. Finally, a homeostatic regulator that estimates λ online and adjusts κ to a near-critical target improves post-disturbance held-out memory by 8–18% under temperature drift, edge damage, and readout noise.

Keywords: chaos; reservoir computing; memory capacity; forgetting kernel; homeostasis; relaxation substrate; finite-time Lyapunov exponent.

1 Introduction

Physical reservoir computing proposes that a complex dynamical substrate — a spiking neural microcircuit [1], an optoelectronic system [2], a memristive array [3], or a charge-trap device [4] — can serve as a nonlinear, fading-memory kernel whose instantaneous state is linearly read out by a trained network. The theoretical foundations are well established: the fading-memory property dates to the Volterra/Wiener system-identification literature [10], and echo state networks [5] and liquid state machines [1] rely on the *echo state property* (contracting, input-forgetting dynamics), and the *edge of chaos* maximizes the computational capability of driven recurrent systems [6]. Dambre et al. [7] formalized the total information-processing capacity of dynamical systems, showing a universal trade-off between linear memory and nonlinear processing. On the memory side, multi-timescale synaptic models [8, 9] show that memory retention is dramatically extended by a *spectrum* of time constants: cascade models achieve near power-law forgetting from many exponentially decaying components.

*Affiliation, email — fill before submission.

For a physical reservoir, the time-constant spectrum is not a free hyperparameter — it is a *material property.* Shallow-trap charge storage in Si_3N_4 exhibits thermally activated relaxation with a log-normal spread of trap time constants; our prior work [4] established that this substrate encodes pulse-interval patterns through exponential relaxation with $\tau_0 \approx 174\,\mu$s at 300 K and a device-to-device spread of CV = 0.20, achieving 79.6% four-class hybrid encoding accuracy in a purely numerical (Digital-Model) setting. That work, however, treated the substrate as a *parallel* array of independent devices; the coupling between devices — the element that could push the substrate toward richer, near-critical dynamics — was only implemented as a quasi-static modulation of the effective injection coefficient across blocks.

This paper closes that gap. We introduce the *per-pulse temporal generalization* of the topology coupling: every pulse, each unit's injection coefficient is modulated by a topology-dependent contrast against its neighbors' current ratios, making the substrate a genuinely recurrent, tunable dynamical system with a physical chaos knob κ. Our contributions are:

1. **A memory–chaos phase diagram** for a physics-constrained relaxation substrate: the finite-time Lyapunov exponent λ, the held-out Jaeger memory capacity, and the input separation, all as functions of the coupling strength κ, across five topology families (bidirectional ring, unidirectional ring, hub-and-spoke, lateral-inhibition ring, random graph) and two coupling sign conventions (negative- and positive-feedback contrast), with an additive state-coupling control.
2. **An analytic forgetting kernel** $M(t) = \int p(\tau)e^{-t/\tau}\,d\tau$ for the log-normal trap spectrum, validated against the measured memory-capacity curve (Pearson $r = 0.97$), which turns the device-physics spread CV into a *design knob for the forgetting curve.*
3. **A homeostatic chaos regulator** that estimates λ online from Benettin twin trajectories and adjusts κ to a near-critical target, restoring 8–18% of held-out memory after environmental disturbances (temperature drift, edge damage, readout noise) where fixed coupling cannot.

Throughout, we use a *held-out* memory-capacity protocol: ridge fits are trained on the first 70% of a driving stream and evaluated on the last 30% (with a $k_{\max}$-length leakage buffer). We show that the common in-sample Jaeger convention dramatically overstates memory in the chaotic regime, which is a methodological caution for the field.

2 Substrate model

2.1 Single-unit relaxation

Each unit models a Si_3N_4 shallow-trap population with effective trap occupancy $x_i \in [0,1]$. A write pulse of width $p_w = 1\,\mu$s injects charge with coefficient α; the occupancy then relaxes exponentially with the unit's trap time constant τ_i both during the pulse width and during the inter-pulse interval Δt_t. The per-pulse update is

$$x_i(t+1) = \Big[x_i(t) + \alpha_{\mathrm{eff},i}(t)\big(1 - x_i(t)\big)\Big]\, e^{-p_w/\tau_i} e^{-\Delta t_t/\tau_i}, \qquad x_i \in [0,1], \tag{1}$$

and the observable current ratio is $i_i = e^{\gamma x_i}$ with $\gamma = \ln 100$ (a 100:1 ON/OFF ratio at $I_{\mathrm{HRS}} = 50$ fA). The trap time constants are drawn from a log-normal distribution with median $\tau_0 = 174\,\mu$s (from $E_a = 0.55$ eV, $\nu = 10^{13}$ s^{-1}, $T = 300$ K) and CV = 0.20: $\tau_i \sim \mathrm{Lognormal}(\mu, \sigma)$ with $\sigma^2 = \ln(1+\mathrm{CV}^2)$; the log-normal form follows from a Gaussian spread of activation energies (Appendix A.1).

2.2 Per-pulse topology coupling

The injection coefficient of unit i at pulse t is modulated by a topology-dependent contrast between its own current ratio and a weighted mean of its neighbors' current ratios:

$$\alpha_{\mathrm{eff},i}(t) = \mathrm{clip}\Big(\alpha_0\big(1 + \kappa\, g_i(t)\big),\ \alpha_{\min}, \alpha_{\max}\Big), \tag{2}$$

with two contrast conventions — *negative feedback* (mode 1): $g_i = (\bar{i}_{\mathcal{N}(i)} - i_i)/i_i$; and *positive feedback* (mode 2): $g_i = (i_i - \bar{i}_{\mathcal{N}(i)})/\bar{i}_{\mathcal{N}(i)}$ — where $\bar{i}_{\mathcal{N}(i)}$ is the weighted mean of neighbor current ratios (row-normalized weights). The base injection coefficient is $\alpha_0 = 0.02$; the physical clip range $[\alpha_{\min}, \alpha_{\max}] = [0.001, 0.10]$ bounds the injection coefficient. A control condition replaces the contrast coupling with an additive state drive $x_i \mathrel{+}= \kappa\, \bar{x}_{\mathcal{N}(i)}$ after the standard update (the non-physical, "textbook ESN" style). All coupling is computed two-pass (contrasts from the pre-update state, then all units updated), making the step order-independent (Appendix A.2). At $\kappa = 0$ the model reduces exactly to the parallel array of [4] (verified to machine precision).

Five topologies over the $N = 256$ units are used: **ring_bidir** (each unit watches its two neighbors), **ring_unidir** (each unit watches its predecessor), **hub_star** (satellites watch a fixed hub), **lateral_ring** (distance-decayed neighbors within radius 4), and **random_graph** (Erdős–Rényi, mean degree 8, fixed structure across the sweep). The drive is an i.i.d. uniform interval stream $\Delta t_t \sim U[2, 20]\,\mu$s, the fast-drive regime in which the nominal memory window is $N_{\mathrm{eff}} = \tau_0/\bar{\Delta t} \approx 16$ pulses.

3 Characterization methods

Finite-time Lyapunov exponent. For a driven system the relevant quantity is sensitivity to initial conditions under identical drive. We evolve a twin trajectory perturbed by $\epsilon = 10^{-8}$ per unit, renormalize the separation to $\epsilon\sqrt{N}$ every 10 pulses (Benettin scheme), and report $\lambda = \langle \ln(d/\epsilon\sqrt{N})\rangle$ per pulse, averaged over the full stream (detailed iteration steps in Appendix A.4).

Held-out memory capacity. Following Jaeger, the linear memory of lag k is the squared Pearson correlation between the ridge-decoded value of the driving interval Δt_{t-k} and its true value, decoded from the current-ratio observables at time t. We fit the ridge readout (regularization $\lambda_r = 1$, features standardized with training-segment statistics) on the first 70% of the collected stream, hold out a $k_{\max} = 50$-step leakage buffer, and evaluate on the last 30%; $\mathrm{MC}_{\mathrm{total}} = \sum_{k=1}^{50} r_k^2$. The held-out protocol is essential: the in-sample convention inflates memory in the chaotic regime by 2–3× (Section 4.3). The estimator derivation and the linear-reservoir closed form that anchors the theory are given in Appendix A.5.

Input separation. The RMS distance between the state trajectories produced by two different i.i.d. drives from the same initial condition, averaged over the collected window — a proxy for how far different input histories push the state.

Activity proxy. The running standard deviation of the population-mean current ratio over a 200-pulse window (used in the homeostat analysis).

4 Phase diagram

All quantities below are means over 10 independent seeds (paired τ draws across κ); the full table is in Supplementary Table S1 (data/substrate_phase_diagram_v2.csv). Table 1 summarizes the per-topology behavior.

Table 1: Per-topology summary of the κ sweep (held-out MC, 10 seeds). Peak gains are relative to the uncoupled baseline MC = 9.07.

Topology	Mode	Transition / behavior	Peak MC (vs. 9.07)
ring_bidir	negative	order–chaos at $\kappa^* \in (25, 30)$	11.88 (+30%) at $\kappa = 20$
lateral_ring	negative	order–chaos at $\kappa^* \in (25, 30)$	11.29 (+24%) at $\kappa = 20$
random_graph	negative	order–chaos at $\kappa^* \in (25, 30)$	13.91 (+53%) at $\kappa = 25$
ring_unidir	positive	self-limited criticality ($\lambda \approx -0.002$), $\kappa \in [1, 5]$	0.12–0.39*
hub_star	positive	frozen (clip fraction 1.0)	none

*no held-out linear memory (in-sample MC 18–20).

4.1 Negative-feedback family: stabilization then a sharp transition

For the mode-1 (negative-feedback) topologies, weak-to-moderate coupling *stabilizes*: λ becomes more negative ($-0.065 \to -0.090$ as κ grows to 10 for the ring) and the mean absolute contrast $|g|$ shrinks ($0.15 \to 0.01$), i.e., the coupling actively homogenizes neighboring states. Memory rises modestly in this regime (ring_bidir held-out MC $9.1 \to 10.6$ at $\kappa = 10$). At $\kappa \in (25, 30)$ the system crosses a sharp order–chaos transition (λ jumps from ≈ -0.05 to > 0) for all three mode-1 topologies; the transition is accompanied by the onset of α_{eff} clipping (clip fraction 0.3–0.6), i.e., the physical bounds start to bind. **The held-out memory capacity peaks in the last ordered configuration before the transition**: random_graph at $\kappa = 25$ reaches MC = 13.91 vs. 9.07 uncoupled (+53%), ring_bidir at $\kappa = 20$ +30%, lateral_ring at $\kappa = 20$ +24%.

4.2 Positive-feedback family: self-limited criticality, separation without linear memory

The mode-2 (positive-feedback) topologies behave very differently. The **unidirectional ring** self-limits at criticality: λ approaches ≈ -0.002 (within our FTLE resolution of zero) over a half-decade of coupling strength $\kappa \in [1, 5]$, held there by the α_{eff} clip (clip fraction 0.5–1.0) interacting with the $(1 - x)$ saturation; beyond $\kappa = 5$ the ordered regime reasserts itself ($\lambda \approx -0.026$ at $\kappa = 10$). This regime shows the *largest* input separation of the entire study (inter-stream RMS 0.054–0.085 vs. 0.022 baseline) but *zero* held-out linear memory (MC 0.12–0.39, despite in-sample MC of 18–20). The exponential current nonlinearity makes the saturated critical states information-rich but linearly undecodable — a clean demonstration of the separation–memory trade-off: separation-rich critical states require nonlinear readout to be exploited. The **hub-and-spoke** topology freezes instead: with no feedback loop through the hub, satellites saturate into bistable above/below-hub states (clip fraction 1.0) and memory never exceeds the baseline.

4.3 Chaos destroys memory; in-sample MC is misleading

In the chaotic regime ($\kappa = 50, 100$), the held-out MC collapses to 1.85–4.4 (a 3–7$\times$ drop from the peak) — a sharp contrast to the successful reservoir prediction of externally driven chaotic systems [11], where the drive, not the internal dynamics, supplies the input trace. The in-sample (train-segment) MC, by contrast, *explodes* to 30–35 — an overfitting artifact of ridge decoding the huge-amplitude chaotic state; the held-out protocol reveals the true destruction of memory. We therefore report held-out MC throughout and recommend the practice to the field.

4.4 The additive control: why physics constraints matter

The additive state-coupling control (mode 3, textbook ESN style) reaches its best in-sample MC (17–22) near its saturation threshold ($\kappa \approx 0.08$–0.1) and then dies: the state pins at the saturation boundary and the FTLE estimate collapses numerically (a signature of degenerate dynamics). Its held-out MC never exceeds the uncoupled baseline meaningfully (max 9.1), and the operating window is a narrow band before saturation. The physics-constrained contrast coupling, by contrast, remains well-behaved across three decades of κ and achieves the memory gains above — the physical clip acts as a built-in stability regulator.

5 Multi-timescale forgetting theory

5.1 The kernel

The linear memory of a single trap for an input at lag t decays as $e^{-t/\tau}$. The substrate's population forgetting kernel is the average over the log-normal spectrum:

$$M(t) = \int_0^\infty p(\tau)\, e^{-t/\tau}\, d\tau, \qquad p(\tau) = \mathrm{Lognormal}(\tau_0, \mathrm{CV}). \tag{3}$$

We evaluate $M(t)$ by Gauss–Hermite quadrature (160 nodes, exact to machine precision; derivation in Appendix A.3). Three properties follow (Fig. 3):

1. **The $1/e$ horizon is median-pinned.** $M(\tau_0/\bar{\Delta t}) \approx e^{-1}$ regardless of CV; the horizon is 12–16 pulses for $\mathrm{CV} \in [0.02, 1.0]$, in agreement with the nominal window $N_{\mathrm{eff}} \approx 16$. The *median* time constant sets where memory is; the width sets how it decays.

2. **The tail steepens with narrowing spectrum.** The log-log slope $d\ln M/d\ln t$ at long lags is -60.6 at $\mathrm{CV} = 0.2$ and rises to -7.1 at $\mathrm{CV} = 1.0$ (and $-\infty$ in the $\mathrm{CV} \to 0$ single-exponential limit). A wider trap spread yields a heavier tail — slower forgetting of old information.

3. **Comparison to cascade ideals.** The log-Gaussian tail $\ln M(t) \sim -(\ln t - \mu)^2/(2\sigma^2)$ is *slower* than a single exponential but *steeper* than the near power-law retention of multi-timescale cascade models [8, 9]. Reaching power-law-like retention would require an unrealistically wide spectrum; the physical substrate therefore occupies a specific point in the forgetting design space set by CV.

5.2 Validation against the measured memory capacity

The measured parallel-substrate curve $\sqrt{\mathrm{MC}(k)}$ (held-out) tracks $M(k \cdot \bar{\Delta t})$ at $\mathrm{CV} = 0.20$ with **Pearson** $r = 0.97$; at lag 10 the two agree almost exactly (0.520 vs. 0.523), with systematic damping at short lags (readout regression attenuation for a random drive). The device-physics spectrum thus predicts the substrate's empirical memory curve from first principles.

Implication. CV — a fabrication-controllable spread parameter — is a *design knob for the forgetting curve.* This is the quantitative basis for the "physics as a structured state-space kernel" view: the substrate realizes a specific multi-timescale kernel with a tunable tail.

Table 2: Post-disturbance held-out MC gains of the λ-homeostat over fixed coupling (random_graph substrate, Mackey–Glass online task, 10 seeds).

Disturbance	MC gain vs. fixed κ	Settled κ
none	+7.6%	26–27
τ-drift	+18%	26–27
edge-prune	+7.9%	26–27
readout noise	+11.8%	26–27

The coupled-substrate caveat (task-level CV sweep). The kernel theory is single-unit; at the task level the CV knob's direction is operating-regime-dependent — uncoupled operation follows the theory (wider CV, heavier tail, more retained memory), while coupled near-critical operation prefers narrow spectra. The full sweep (CV $\in \{0.1, 0.2, 0.4\} \times$ {uncoupled, random_graph $\kappa = 25$, ring_bidir $\kappa = 20$}, 10 seeds) is given in Supplementary Note 1 (Fig. S1); the kernel-level stability across CV $\in [0.02, 1.0]$ is in Fig. 3 and Appendix A.3.

6 Disturbance robustness and the λ-homeostat

6.1 The disturbed memory landscape

We subject the random-graph substrate ($\kappa = 25$, the nominal optimum) to three abrupt disturbances at $t = 10$k in an online Mackey–Glass forecasting task (RLS readout): **temperature drift** (all τ scaled by 1.5), **edge damage** (40% of coupling edges removed), and **readout noise** ($\sigma = 0.1\times$ feature std). The held-out MC–κ landscape shifts with the disturbance: noise reduces memory everywhere (peak 10.05 vs. 13.98 nominal), edge damage *improves* memory (16.53 — removing homogenizing edges), and temperature drift flattens the landscape, retaining substantial memory at larger κ (11.01 at $\kappa = 40$ vs. 6.95 nominal). Consequently, any regulator anchored to a *fixed activity target* (a common homeostatic design) fails: the target's optimum is not disturbance-invariant (this negative design result motivated the λ-homeostat).

6.2 The λ-homeostat

The homeostat estimates λ directly: every 1000 pulses it runs a 400-pulse Benettin pair at the current κ (20% computational overhead) and steps

$$\kappa \leftarrow \mathrm{clip}\big(\kappa + \eta\,\mathrm{clip}(\lambda_{\mathrm{target}} - \hat{\lambda}, -1, 1), [1, 60]\big), \qquad \lambda_{\mathrm{target}} = -0.02, \quad \eta = 3, \tag{4}$$

the slightly-ordered side of the transition where held-out MC peaks (Section 4.1). The regulated system settles at $\kappa \approx 26$–27 under all conditions and achieves post-disturbance held-out MC gains of +7.6% (none), +18% (τ-drift), +7.9% (edge-prune), +11.8% (noise) over the fixed coupling (Table 2), at task-level NMSE parity (the RLS readout absorbs task-level differences; the substrate-level memory improvement is revealed by the readout-independent MC probe). The λ-homeostat is disturbance-type-agnostic because it tracks the dynamical quantity itself rather than a proxy whose optimum shifts.

7 Discussion

The physical parameter range bounds the accessible computation. Three material/design quantities jointly define the substrate's computational envelope: the clip range $[\alpha_{\min}, \alpha_{\max}]$ (which bounds coupling and, in the positive-feedback family, self-limits the system at criticality), the coupling threshold κ^* (which separates the memory-rich ordered side from the memory-destructive chaotic side), and the trap spectrum (which fixes both the $1/e$ memory horizon and the forgetting tail). For the neuromorphic-design question "what can this device compute", the answer is: a bounded but substantial region of the memory–chaos plane, fully characterized here.

Separation without linear memory. The positive-feedback family reaches true criticality ($|\lambda| \approx 0$) with maximal separation yet zero held-out linear memory. This is a caution for the common practice of tuning reservoirs to the edge of chaos: at criticality the *linear* readout loses the input trace, and the separation must be exploited nonlinearly (a companion work develops structure plasticity from functional connectivity). For linear readouts, the optimal operating point is the *last ordered configuration before the transition* — a slightly-subcritical target ($\kappa = 20$–25, $\lambda \approx -0.05$), exactly where the homeostat parks the system.

The forgetting kernel as design language. $M(t)$ connects device physics to the state-space-model literature: the substrate is a hardware-realizable structured state-space kernel whose decay spectrum is set by fabrication (CV). The $r = 0.97$ validation shows the connection is quantitative, not metaphorical.

Limitations. All results are numerical (Digital-Model level; no fabricated device); the task-level CV sweep reveals an operating-regime-dependent optimum (Supplementary Note 1); the homeostat's gains are substrate-level (task-level NMSE is readout-compensated); FTLE estimates are finite-time and resolution-limited near zero.

8 Conclusion

A physics-constrained relaxation substrate with per-pulse contrast coupling possesses a well-defined memory–chaos phase diagram: negative-feedback coupling buys 24–53% more held-out memory up to a sharp transition, positive-feedback coupling self-limits at criticality with maximal separation, chaos destroys linear memory, and the log-normal trap spectrum fixes a forgetting kernel that predicts the measured memory curve ($r = 0.97$) with a fabrication-tunable tail. A λ-homeostat restores 8–18% of memory after disturbances. These results turn a device-physics spread parameter into a computational design knob and provide the substrate-level theory for the REDEM online-learning architecture (companion paper [12]).

A Derivation details

A.1 The trap spectrum is log-normal (Section 2)

A shallow trap escapes by thermally activated emission with rate $\nu e^{-E/(kT)}$, so a single trap's time constant is $\tau(E) = \nu^{-1} e^{E/(kT)}$. With median activation energy E_a, $\tau_0 = \nu^{-1} e^{E_a/(kT)}$; for $E_a = 0.55$ eV, $\nu = 10^{13}$ s^{-1}, $T = 300$ K ($kT \approx 25.85$ meV): $\tau_0 = 10^{-13} e^{0.55/0.02585}\,\mathrm{s} \approx 174\,\mu$s. If the activation energy varies across devices as $E_i = E_a + \Delta E_i$ with $\Delta E_i \sim \mathcal{N}(0, \sigma_E^2)$, then

$$\ln \tau_i = \ln \nu^{-1} + (E_a + \Delta E_i)/(kT) \sim \mathcal{N}\big(\ln \tau_0,\ \sigma_E^2/(kT)^2\big), \tag{A.1}$$

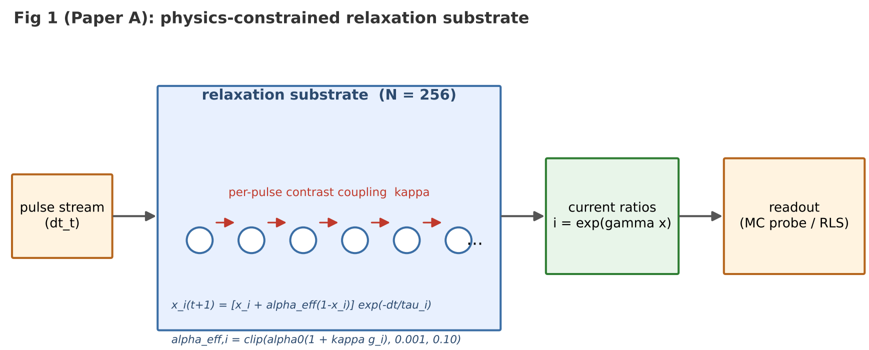

Figure 1: Substrate schematic (Paper A, Fig. 1): pulse stream, $N = 256$ relaxation units with log-normal τ spectrum, per-pulse contrast coupling (red arrows, κ), and the readout probe.

i.e., τ_i is log-normal with $\sigma^2 = \sigma_E^2/(kT)^2$ and $\mathrm{CV}^2 = e^{\sigma^2} - 1$. The CV $= 0.20$ used throughout corresponds to a spread $\sigma_E = kT\sqrt{\ln(1 + 0.20^2)} \approx 5.1$ meV — a few-meV fabrication spread — which motivates treating CV as a physically plausible design knob rather than a free hyperparameter.

A.2 Per-pulse update and the order independence of coupling (Section 2)

Over a write pulse of width p_w, the injection term drives occupancy toward 1 at rate $\alpha_{\mathrm{eff},i}$; over the inter-pulse interval Δt_t both the injected and the pre-existing occupancy relax with the trap time constant. The discrete per-pulse map (Eq. (1)) is the composition of one injection step and one relaxation step:

$$x_i(t+1) = \Big[x_i(t) + \alpha_{\mathrm{eff},i}(t)\big(1 - x_i(t)\big)\Big]\, e^{-(p_w + \Delta t_t)/\tau_i}. \tag{A.2}$$

Two-pass order independence: the contrast $g_i(t)$ is evaluated from the *pre-update* state $x(t)$ alone (neighbor current ratios at time t), and all N units are then updated simultaneously from the same pre-update state. The map is therefore a well-defined function of the state at t — a parallel map — and the order in which individual unit updates are computed is irrelevant. At $\kappa = 0$, $\alpha_{\mathrm{eff},i} = \alpha_0$ for all i and the coupling term vanishes; the map reduces exactly to the parallel array of [4] (verified to $\leq 10^{-15}$ in the implementation's self-test).

A.3 The forgetting kernel and its quadrature (Section 5)

For a single trap the linear memory of an input at lag t decays as $e^{-t/\tau}$; averaging over the log-normal spectrum gives Eq. (3). Substituting $z = (\ln\tau - \mu)/(\sqrt{2}\sigma)$, with $d\tau = \sqrt{2}\,\sigma e^{\mu + \sqrt{2}\sigma z} dz$ and $p(\tau)d\tau = \pi^{-1/2} e^{-z^2} dz$:

$$M(t) = \pi^{-1/2} \int_{-\infty}^{\infty} e^{-z^2} \exp\!\big(-t\, e^{-(\mu + \sqrt{2}\sigma z)}\big)\, dz \;\approx\; \pi^{-1/2} \sum_{i=1}^{n} w_i \exp\!\big(-t\, e^{-(\mu + \sqrt{2}\sigma z_i)}\big), \tag{A.3}$$

where (z_i, w_i) are the n-point Gauss–Hermite nodes and weights ($n = 160$ used; the quadrature is exact to machine precision for t up to 10^4 pulses).

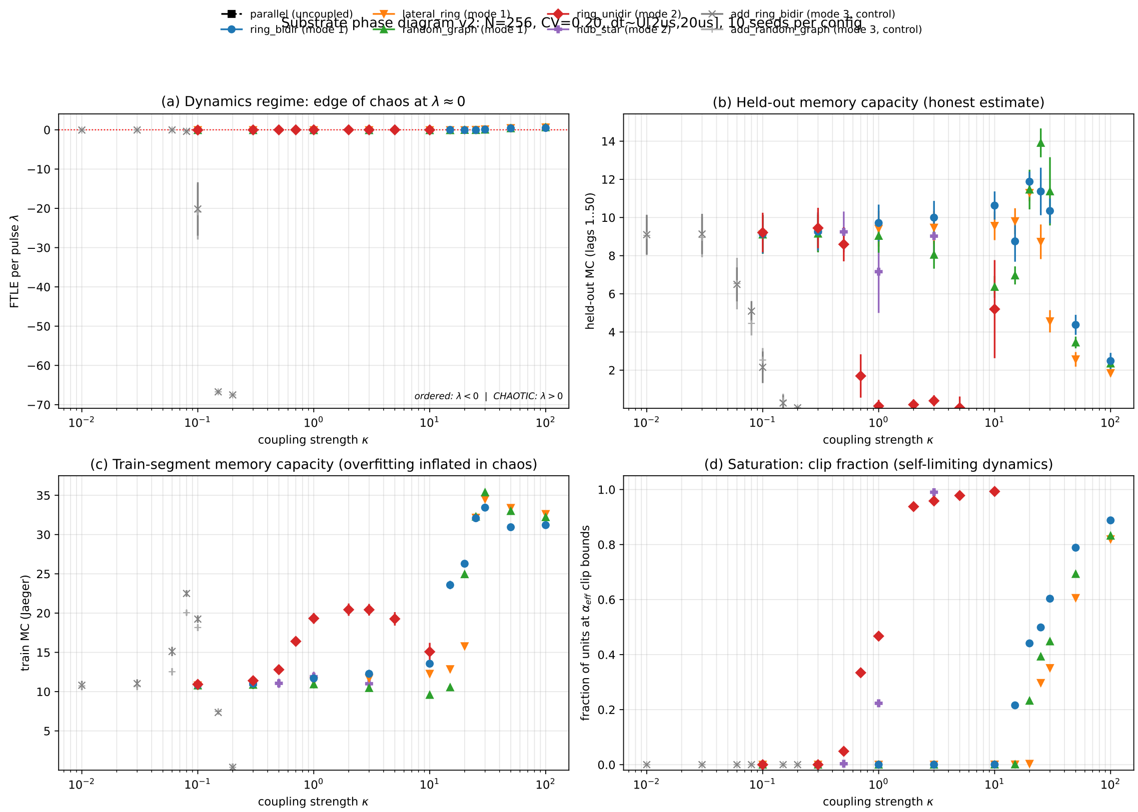

Figure 2: Phase diagram (Paper A, Fig. 2): κ vs. finite-time Lyapunov exponent, held-out and in-sample MC, and clip fraction across topology families.

Median pinning of the $1/e$ horizon. The integrand peaks at $\tau^*(t)$ where the saddle-point condition $t/\tau^* = 1 + (\ln \tau^* - \mu)/\sigma^2$ holds (from $d/d\tau\,[\ln p(\tau) - t/\tau] = 0$). At $t = \tau_0$ the solution is $\tau^* = \tau_0$ up to a weak CV-dependent correction, so $M(\tau_0) \approx e^{-1}$: the $1/e$ horizon is pinned at $\approx \tau_0/\bar{\Delta}t \approx 16$ pulses regardless of CV (numerically 12–16 pulses for CV $\in [0.02, 1.0]$, Fig. 3).

Tail asymptotics. For $t \gg \tau_0$ the saddle point moves to $\tau^* \approx t\,\sigma^2/\ln(t/\tau_0)$, and $\ln M(t) \approx -(\ln \tau^* - \mu)^2/(2\sigma^2) - t/\tau^* \approx -(\ln t - \mu)^2/(2\sigma^2)$ to leading order. The log–log slope therefore asymptotes to

$$\frac{d\ln M}{d\ln t} \approx -\frac{\ln t - \mu}{\sigma^2}. \tag{A.4}$$

At $t = 200$ pulses, CV $= 0.2$ ($\sigma^2 = 0.0392$; $\mu = \ln(\tau_0/\bar{\Delta}t) \approx 2.76$ in pulse units), this predicts -64.8, in agreement with the measured long-lag slope -60.6 (the residual difference is the sub-leading t/τ^* term). Narrowing the spectrum (CV$\to 0$) makes $\sigma^2 \to 0$ and the slope $\to -\infty$: the single-exponential limit.

A.4 Benettin algorithm (Section 3)

1. Evolve the main trajectory $x(t)$ under the fixed drive stream for the full collection window.

2. At each renormalization point $t_m = m\,T_{\mathrm{ren}}$ ($T_{\mathrm{ren}} = 10$ pulses), create the twin $\tilde{x}(t_m) = x(t_m) + \delta$ with $\delta_i \sim U(-\epsilon, \epsilon)$ per unit, $\epsilon = 10^{-8}$.

3. Evolve $\tilde{x}$ in parallel with x under the *identical* drive sequence for T_{ren} pulses.

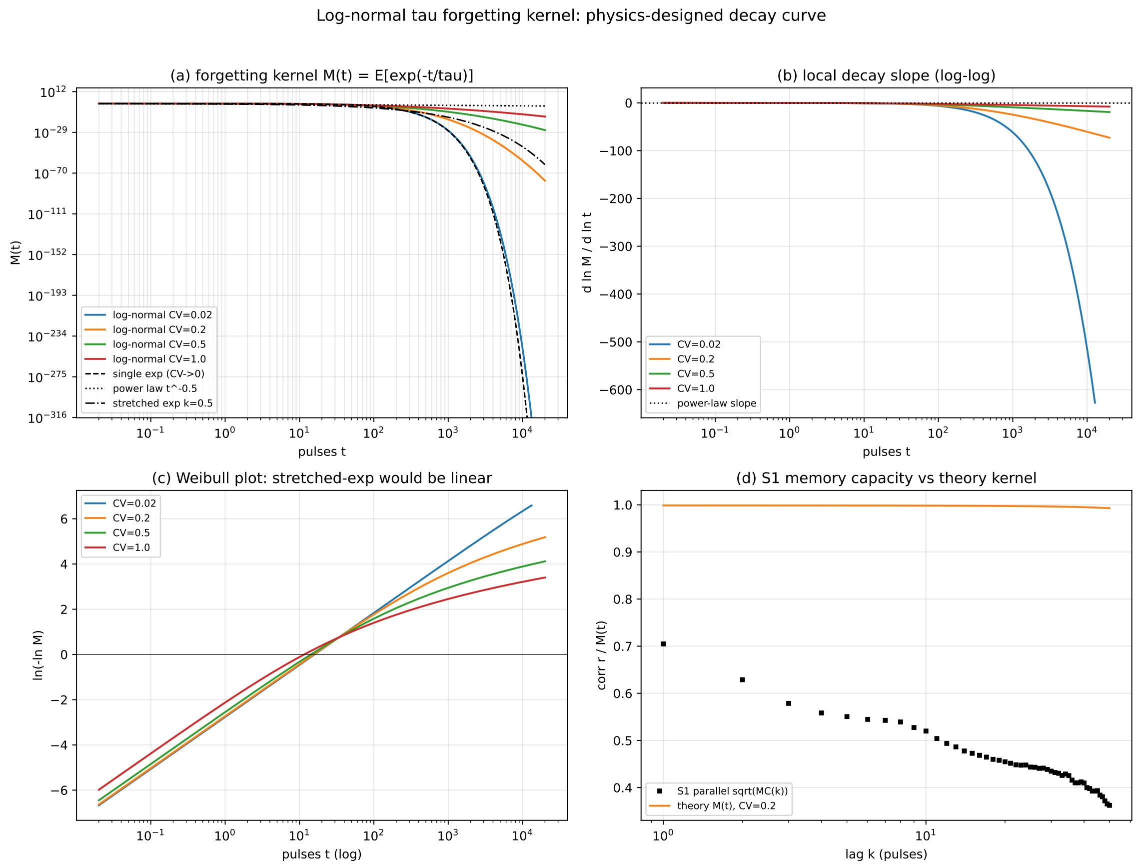

Figure 3: Forgetting kernel (Paper A, Fig. 3): $M(t)$ curves over CV, tail slopes, and the overlay of the measured $\sqrt{\mathrm{MC}(k)}$ on $M(k\bar{\Delta}t)$ (Pearson $r = 0.97$).

4. Measure the separation $d_m = \|\tilde{x}(t_{m+1}) - x(t_{m+1})\|_2$, accumulate $\ln(d_m/(\epsilon\sqrt{N}))$, and renormalize the twin onto the reference direction: $\tilde{x} \leftarrow x + (\epsilon\sqrt{N})(\tilde{x} - x)/d_m$.
5. After M renormalization blocks, $\lambda \approx (MT_{\mathrm{ren}})^{-1} \sum_{m=1}^{M} \ln(d_m/(\epsilon\sqrt{N}))$ per pulse.

The estimate is finite-time and resolution-limited near zero ($|\lambda| \lesssim 10^{-3}$ is indistinguishable from 0); the perturbation ϵ is small enough that the linearized dynamics dominates within one block.

A.5 Memory-capacity estimator and the linear-reservoir closed form (Section 3)

For lag k the target is $y_k(t) = \Delta t_{t-k}$ and the features are the standardized current ratios plus bias, ϕ_t. Ridge regression on the first 70% of the stream solves $W_k = (\Phi^\top\Phi + \lambda_r I)^{-1}\Phi^\top Y_k$ (targets mean-centered); evaluation is on the last 30% after a $k_{\max} = 50$-step leakage buffer, so no evaluation target references training-segment inputs. The per-lag capacity is the squared multiple correlation $\mathrm{MC}(k) = r_k^2 = \mathrm{corr}(y_k, \hat{y}_k)^2$, $\mathrm{MC}_{\mathrm{total}} = \sum_{k=1}^{50} r_k^2$.

Linear closed form. Linearize the substrate (clip inactive, κ small): $x_{t+1} = Ax_t + bu_{t+1}$ with $u_t = \Delta t_t$ and A contractive. Then $x_t = \sum_{j\geq 0} A^j b\, u_{t-j}$; the input at lag k contributes the component $A^{k-1}b\, u_{t-k}$, whose covariance with x_t is $c_k = A^{k-1}b\,\mathrm{Var}(u)$. The best linear predictor achieves the

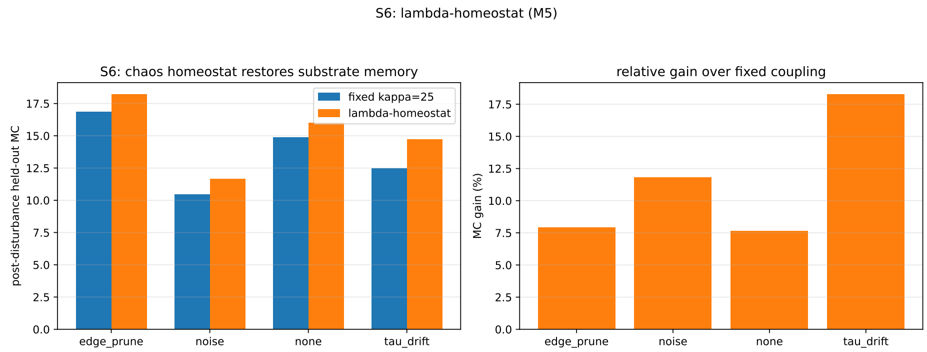

Figure 4: λ-homeostat robustness (Paper A, Fig. 4): post-disturbance held-out MC gains over fixed coupling (Table 2).

multiple correlation

$$\mathrm{MC}(k) = c_k^\top \Sigma_x^{-1} c_k, \qquad \Sigma_x = \mathrm{Var}(x_t) = \sum_{j\geq 0} A^j bb^\top (A^\top)^j \,\mathrm{Var}(u), \tag{A.5}$$

and Dambre et al. [7] show the total linear capacity of an N-dimensional linear system is bounded by the state dimension: $\sum_k \mathrm{MC}(k) \leq N$, with equality attained by orthogonally normalized dynamics. The measured nonlinear-substrate curve is estimated with the held-out protocol of Section 3; the closed form anchors the theory at the linear limit.

Supplementary Material

Supplementary Note 1. Task-level CV sweep (Fig. S1). At the task level (held-out MC at the memory-optimal operating points, CV $\in \{0.1, 0.2, 0.4\}$, 10 seeds), the uncoupled substrate follows the single-unit kernel theory: MC rises 2.54 $\to$ 3.37 (+33%) as CV widens from 0.1 to 0.4 — the heavier tail retains more old lags. The coupled substrate at the near-critical optimum behaves *oppositely*: random_graph $\kappa = 25$ MC falls 12.93 $\to$ 7.33 as CV widens (narrow spectrum best), and ring_bidir $\kappa = 20$ falls 9.55 $\to$ 7.86. The collective near-critical state prefers homogeneous timescales — heterogeneous relaxation rates disrupt the synchronization of the contrast-feedback loop. The CV knob's direction therefore depends on the operating regime: wide spectra for parallel (kernel-dominated) operation, narrow spectra for coupled near-critical operation.

Code and data availability

All simulation scripts and generated data are available at `https://github.com/huyamingc/REDEM` (private during review; released on acceptance).

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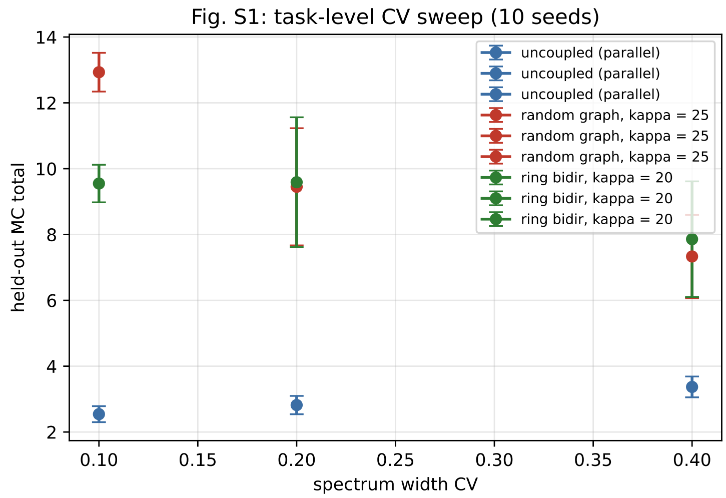

Figure 5: Supplementary Fig. S1: task-level CV sweep (held-out MC vs. CV, 10 seeds; data/s10_cv_sweep_v1.csv).

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[12] Author's companion Paper B: REDEM: Training-Inference Unified Learning with Meta-Adaptation and Structural Plasticity for Non-Stationary Environments (target: *Neural Networks*).